```
%task#1 %%%%%%
```

```
clc;
clear;
close all;

K = 2;
tau = 5;

G = tf(K, [tau 1]);

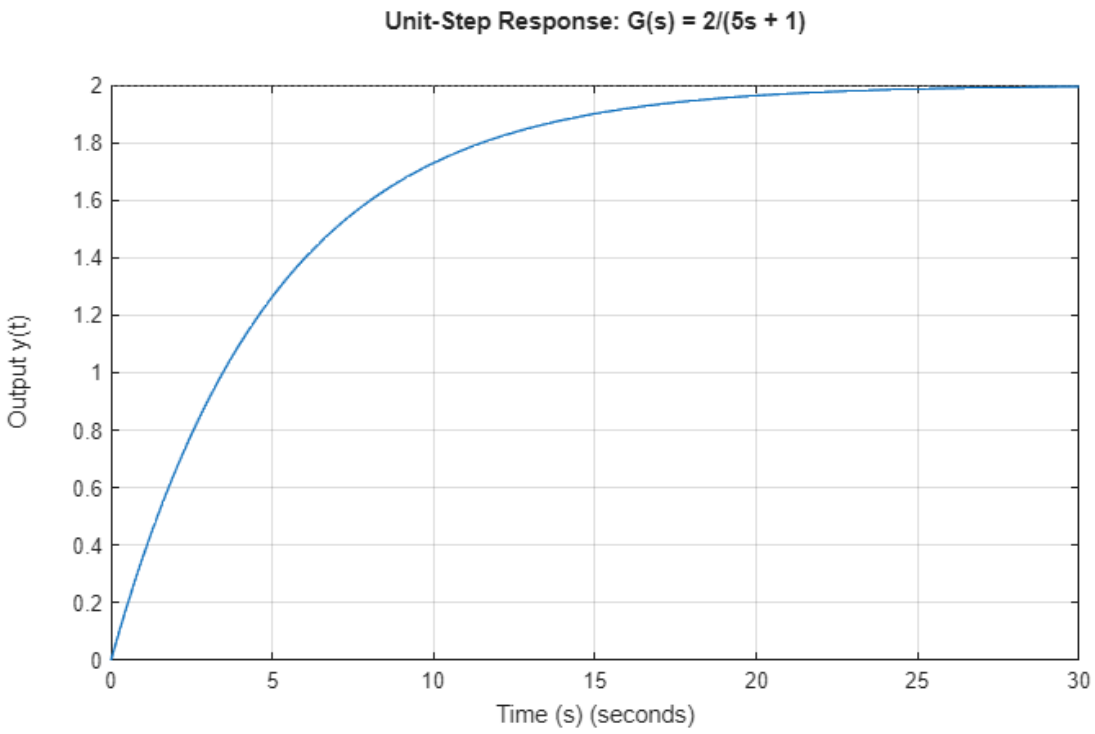
figure;
step(G);
grid on;

title('Unit-Step Response: G(s) = 2/(5s + 1)');
xlabel('Time (s)');
ylabel('Output y(t)');

system_pole = pole(G)
steady_state_gain = dcgain(G)
response_info = stepinfo(G)
```

```
system_pole =
   -0.2000
steady_state_gain =
     2
response_info =
  struct with fields:

         RiseTime: 10.9850
    TransientTime: 19.5604
     SettlingTime: 19.5604
      SettlingMin: 1.8090
      SettlingMax: 1.9987
        Overshoot: 0
       Undershoot: 0
             Peak: 1.9987
         PeakTime: 36.6111
```



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